NEWTON SCHOOL

ALGEBRA 1 | ANSWER KEY | v1 · ROUND 1

Solving Quadratics — Method Selection & Solve

Student: Evelyn Yoo **Date:** 2026-05-01 (Fri)

Part A — Method Identification

Problem 1

$2x^2 + 5x - 3 = 0$

$D = 5^2 - 4(2)(-3) = 25 + 24 = 49 = 7^2$ **perfect square ✓**

$b = 5$ (odd), but D is PS — PS check wins.

METHOD: FACTOR

Problem 2

$x^2 + 4x + 1 = 0$

$D = 16 - 4 = 12$ **NOT a perfect square**

$b = 4$ **even**

METHOD: COMPLETE THE SQUARE

Problem 3

$x^2 + 3x - 5 = 0$

$D = 9 + 20 = 29$ **NOT a perfect square**

$b = 3$ **odd**

METHOD: QUADRATIC FORMULA

Problem 4

$x^2 - 8x + 12 = 0$

$D = 64 - 48 = 16 = 4^2$ **perfect square ✓**

$b = -8$ even, but PS check wins → factor.

METHOD: FACTOR

Problem 5

$3x^2 + 4x + 1 = 0$

$D = 16 - 12 = 4 = 2^2$ **perfect square ✓**

$b = 4$ even, but PS check wins.

METHOD: FACTOR

Part B — Solve Completely

Problem 6

$x^2 + 6x + 4 = 0$

$D = 36 - 16 = 20$ (not PS), $b = 6$ even $\longrightarrow$ **complete the square**.

Magic number $= (6/2)^2 = 9$.

$x^2 + 6x + 9 - 9 + 4 = 0 \;\Rightarrow\; (x+3)^2 - 5 = 0$

$(x+3)^2 = 5 \;\Rightarrow\; x + 3 = \pm\sqrt{5}$

Complete Square | $a = 1$ | $x = -3 \pm \sqrt{5}$

Problem 7

$x^2 - 5x - 14 = 0$

$D = 25 + 56 = 81 = 9^2$ **PS** $\longrightarrow$ **factor**.

Two numbers: product -14, sum $-5 \longrightarrow -7$ and $+2$.

$(x-7)(x+2) = 0$

Factor | PS Discriminant | $x = 7$ or $x = -2$

Problem 8

$2x^2 + 3x - 1 = 0$

$D = 9 + 8 = 17$ (not PS), $b = 3$ odd $\longrightarrow$ **quadratic formula**.

$$x = \frac{-3 \pm \sqrt{17}}{2(2)} = \frac{-3 \pm \sqrt{17}}{4}$$

Quadratic Formula | b odd | $x = \dfrac{-3 \pm \sqrt{17}}{4}$

Problem 9

$x^2 - 4x - 7 = 0$

$D = 16 + 28 = 44$ (not PS), $b = -4$ even $\longrightarrow$ **complete the square**.

Magic number $= (-4/2)^2 = 4$.

$x^2 - 4x + 4 - 4 - 7 = 0 \;\Rightarrow\; (x-2)^2 - 11 = 0$

$(x-2)^2 = 11 \;\Rightarrow\; x - 2 = \pm\sqrt{11}$

Complete Square | b even | $x = 2 \pm \sqrt{11}$

Problem 10

$3x^2 - 5x + 1 = 0$

$D = 25 - 12 = 13$ (not PS), $b = -5$ odd, $a \neq 1 \rightarrow$ **quadratic formula**.

$$x = \frac{-(-5) \pm \sqrt{13}}{2(3)} = \frac{5 \pm \sqrt{13}}{6}$$

Quadratic Formula

b odd

$$\boxed{x = \frac{5 \pm \sqrt{13}}{6}}$$

Part C — Understanding Check

Problem 11 — Why "inside" not "both sides"?

Sample answer: When $a \neq 1$, the bracket has a multiplying it, so anything that comes *out* of the bracket gets multiplied by a. If you instead "add the magic number to both sides," you must add a times the magic number to the right side — but students forget the a and just add the bare magic number. The equation breaks.

Specific mistake: For $3(x^2 + 2x) = 5$, magic $= 1$. Wrong move: $3(x^2 + 2x + 1) = 5 + 1$. Correct: $5 + 3$ on the right (because $a = 3$ multiplies the 1).

Kenny's fix: Add and subtract inside the bracket. Net change is 0, no chance to forget multiplying by a, because the $-$ piece carries a with it as it leaves the bracket.

Conceptual

Problem 12 — b even: complete square or formula?

Complete the square is cleaner.

Why: Half of an even b is a clean integer, so the magic number is an integer (no fractions). The result $(x + b/2)^2 = k$ leads to $x = -b/2 \pm \sqrt{k}$ in one shot.

Formula gives the same answer, but you end up with $\dfrac{-b \pm \sqrt{D}}{2}$ where both numerator pieces are even, forcing you to reduce. Extra arithmetic, more chances to slip.

Conceptual | Method choice